Student Step-by-Step Guide

Information and Computer Studies 2 (Practical) – 036/2

The School of St. Jude – Preparatory NECTA Exam Series 2, October 2023

This guide shows you how to answer every question in the paper, one click at a time. Follow the steps in order. Where a result is given (for example the values in a query or a formula), check that your screen shows the same thing.

Before you start (all questions)

1. Right-click the Desktop → **New** → **Folder**. Name it with your **examination number** (for example `S2911-0045`). Never use your name or school name.
2. Save every file for every question inside this folder.
3. Type your examination number on every page of your work (in a header, footer, or on the page).
4. Press **Ctrl + S** often so you do not lose work.
5. Answer **Question 3** (it is compulsory) and **one** of Question 1 or Question 2.
6. Print your work and check that the printout matches what is saved in your folder.

Contents

• Question 1 – Web design with HTML (Ministry of Tourism pages)

• Question 2 – Database with MS Access (Kimaro Drinking Company)

• Question 3 – Spreadsheet with MS Excel (NexusSphere payroll) – compulsory

Careful: Page 2 of the original paper (the start of Question 1 and part i) and parts (a) and (b) of Question 2 were missing from the copy used. The steps for those parts follow the standard expected answer; follow your own paper's wording if it differs.

Question 1 – Web design with HTML

You will make five files: **Header.html**, **Main.html**, **Aside.html**, **Footer.html** and **index.html** (the layout page that puts the other four together). The body is split into Main (70%) on the left and Aside (30%) on the right. Header, aside and footer have a **powderblue** background.

Step 1 – Get ready

1. Open your exam-number folder.
2. Copy two pictures of tourist sites (for example Kilimanjaro and Kitulo) into the folder. Rename them `kilimanjaro.jpg` and `kitulo.jpg`.
3. Open **Notepad**: click **Start**, type Notepad, press **Enter**.

How to save each HTML file (do this every time)

1. Click **File** → **Save As**.
2. Go to your exam-number folder.
3. Change **Save as type** to **All Files (*.*)**.
4. Type the file name with .html, e.g. `Header.html`.
5. Set **Encoding** to **UTF-8**, then click **Save**.

Careful: If "Save as type" stays on "Text Documents", the file becomes Header.html.txt and will not open as a web page.

Step 2 – Header.html

Type this in a new Notepad file, then save it as **Header.html**:

```
<!DOCTYPE html>
<html>
<head>
  <title>Header</title>
</head>
<body style="background-color: powderblue;">
  <h1 style="text-align: center;">MINISTRY OF TOURISM</h1>
</body>
</html>
```

Tip: "MINISTRY OF TOURISM" must be heading level 1, so it uses **<h1>**.

Step 3 – Main.html



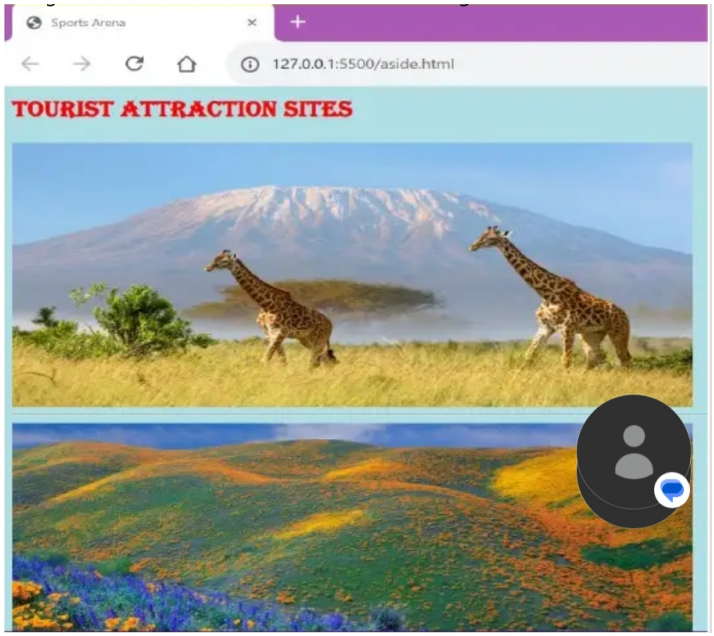
What Main.html should look like

Click **File → New**, type this, then save as **Main.html**:

```
<!DOCTYPE html>
<html>
<head>
  <title>Welcome to our site</title>
</head>
<body style="background-color: yellow;">
  <h2 style="text-align: center;">MY BLESSED COUNTRY, TANZANIA</h2>
  <p style="text-align: center;"><b>This is Tanzania, the country blessed by God Quick Facts.</b></p>
  <ol>
    <li>Tanzania is just a little bit more than twice the size of California.
        Tanzania is the home for Mount Kilimanjaro that is located in Tanzania.
        At 19,341 feet above sea level, it is the largest mountain in Africa.</li>
    <li>Tanzania has the largest concentration of wildlife animals per square
        kilometer, with more than 4 million wild animals and representatives of
        430 species and subspecies. These animals are found in Serengeti,
        Ngorongoro, Mikumi, Kitulo conservation area, Kilimanjaro national parks
        and so many other parts of the country.</li>
  </ol>
</body>
</html>
```

Tip: The paragraphs are numbered 1 and 2 in the picture, so use an ordered list **<ol>** with two **<li>** items, not **<ul>**. "MY BLESSED COUNTRY TANZANIA" must be heading level 2 (**<h2>**).

Step 4 – Aside.html



What Aside.html should look like

Click **File → New**, type this, then save as **Aside.html**:

```
<!DOCTYPE html>
<html>
<head>
  <title>Sports Arena</title>
</head>
<body style="background-color: powderblue;">
  <h2 style="text-align: center; font-family: Algerian; color: red;">TOURIST ATTRACTION SITES</h2>
  <img src="kilimanjaro.jpg" alt="Mount Kilimanjaro" style="width: 100%; height: 100px;">
  <hr>
  <img src="kitulo.jpg" alt="Kitulo Plateau" style="width: 100%; height: 100px;">
</body>
</html>
```

Tip: The heading must be level 2, centred, in the **Algerian** font. Each picture is **width 100%** and **height 100px**. The **<hr>** tag draws the horizontal line between the two pictures.

Step 5 – Footer.html



What Footer.html should look like

Click **File → New**, type this, then save as **Footer.html**:

```
<!DOCTYPE html>
<html>
<head>
  <title>COPYRIGHT</title>
</head>
```

```
<body style="background-color: powderblue;">
  <p style="text-align: center;">&copy; 2023 - ST. JUDE GIRLS SECONDARY SCHOOL.</p>
</body>
</html>
```

Tip: `©` shows the © symbol.

Step 6 – index.html (the layout page)

This page joins the other four pages: header on top, Main (70%) and Aside (30%) side by side, footer at the bottom.

```
<!DOCTYPE html>
<html>
<head>
  <title>Ministry of Tourism</title>
</head>
<frameset rows="20%,65%,15%">
  <frame src="Header.html" name="header">
  <frameset cols="70%,30%">
    <frame src="Main.html" name="main">
    <frame src="Aside.html" name="aside">
  </frameset>
  <frame src="Footer.html" name="footer">
</frameset>
</html>
```

Careful: The 70% / 30% split goes in **cols** (side by side), not rows. File names in src must match your saved names exactly, including capital letters.

Step 7 – Test and print

1. Open your folder and double-click **index.html**. It opens in the browser.
2. Check: header at the top, Main on the left (bigger), Aside on the right, footer at the bottom, and both pictures showing.
3. If a part is blank, check the file name in src and that the file is not saved as .html.txt.
4. Add your exam number at the top of each file as a comment, e.g. `<!-- S2911-0045 -->`, and save again.
5. Print the web page: press **Ctrl + P** → choose the printer → **Print**.
6. If your teacher wants the code printed too: in Notepad click **File** → **Print** for each file.

Question 2 – Database with MS Access

You will make one database with a table called **PRODUCT**, then six saved queries. Menu names below are from Access 2016–2021 / Microsoft 365.

Step 1 – Create the database (part a)

1. Open **Microsoft Access** → click **Blank database**.
2. Click the small folder icon next to File Name → open your exam-number folder.
3. Type a name, e.g. `KIMARO.accdb` → **OK** → **Create**.

Step 2 – Create the PRODUCT table (part b)

1. Click **Create** → **Table Design**.
2. Type these field names and choose these data types:

Field Name	Data Type
PRODUCT NAME	Short Text
PRODUCT CODE	Number
COMPANY NAME	Short Text
UNIT PRICE	Number
DATE OF MANUFACTURE	Date/Time
(second date column in the snip)	Date/Time

1. Press **Ctrl + S**, type `PRODUCT` → **OK**. If Access asks to create a primary key, click **No** (you will set it in Step 3).
2. Click **Home** → **View** → **Datasheet View**.
3. Type all 9 records exactly as shown:

PRODUCT

PRODUCT NAME	PRODUCT CODE	COMPANY NAME	UNIT PRICE	DATE OF MANUFACT	DA
Cocacola	1001	Kwanza	200	2/6/	10/10
Pepsi	1002	Tz Brew	150	3/7/1991	8/12
Kilimanjaro	1003	K Brew	550	5/7/1993	8/22
Tusker	1004	Kibo Brew	550	2/6/1990	10/10
Heinekein	1005	South A	1200	7/9/1999	6/19
Safari	1006	K Brew	600	4/23/1989	6/2
Ndovu	1007	Kwanza	550	8/9/1995	4/26
Kibo gold	1008	South A	550	6/8/1995	10/18
Carlsberg	1009	Kbrew	800	3/12/1996	4/1

The PRODUCT table from the paper

Step 3 – Set the primary key (part c)

1. Click **Home** → **View** → **Design View**.
2. Click the grey box to the left of **PRODUCT CODE** to select that row.
3. Click **Table Design** → **Primary Key**. A small key icon appears.
4. Press **Ctrl + S**.

How to start every query (use this for parts d to i)

1. Click **Create** → **Query Design**.

2. In the Show Table box click **PRODUCT → Add → Close**.
3. Double-click every field in the PRODUCT box so all fields go into the grid at the bottom.
4. Type the criteria or formula (see each part).
5. Click **Run** (the red **!**) to see the result.
6. Press **Ctrl + S**, type the exact query name, click **OK**.

Part (d) – Names beginning with C or ending with A → save as C&A

1. Start a query.
2. In the **Criteria** row under PRODUCT NAME type: `Like "C*" Or Like "*A"`
3. Click **Run**.
4. Save as `C&A`.

Your result should show 2 records: Cocacola and Carlsberg.

Tip: The * means "any letters". "C*" = starts with C; "*A" = ends with A.

Part (e) – Made in 1990 or 1996 → save as 1990 or 1996

1. Start a query.
2. In an empty column's **Field** row type: `YearMade: Year([DATE OF MANUFACTURE])`
3. In the **Criteria** row under it type: `1990 Or 1996`
4. Click **Run**.
5. Save as `1990 or 1996`.

Your result should show: Tusker (2/6/1990) and Carlsberg (3/12/1996).

Part (f) – Codes greater than 1004 and less than 1009 → save as G&L

1. Start a query.
2. In the **Criteria** row under PRODUCT CODE type: `>1004 And <1009`
3. Click **Run**.
4. Save as `G&L`.

Your result should show 4 records: Heinekein 1005, Safari 1006, Ndovu 1007, Kibo gold 1008.

Careful: Do not use "Between 1004 And 1009" — that includes 1004 and 1009, which the question excludes.

Part (g) – Sort lowest to highest Unit Price → save as Highest

1. Start a query.
2. In the **Sort** row under UNIT PRICE choose **Ascending**.
3. Click **Run**.
4. Save as `Highest`.

Order	PRODUCT NAME	UNIT PRICE
1	Pepsi	150
2	Cocacola	200

Order	PRODUCT NAME	UNIT PRICE
3–6	Kilimanjaro, Tusker, Ndovu, Kibo gold	550
7	Safari	600
8	Carlsberg	800
9	Heinekein	1200

Parts (h) and (i) – VAT and Unit Sell Price

(h) Save as KAG&VAT.

1. Start a query with **all** PRODUCT fields.
2. In the next empty column's **Field** row type: `VAT: [UNIT PRICE]*0.04`
3. Click **Run**.
4. Save as `KAG&VAT`.

(i) Save as VAT INCLUSIVE.

1. Start a query with **all** PRODUCT fields.
2. In the next empty column type: `UNIT SELL PRICE: [UNIT PRICE]+[UNIT PRICE]*0.04`
3. Click **Run**.
4. Save as `VAT INCLUSIVE`.

Check your answers:

PRODUCT NAME	UNIT PRICE	(h) VAT	(i) UNIT SELL PRICE
Cocacola	200	8	208
Pepsi	150	6	156
Kilimanjaro	550	22	572
Tusker	550	22	572
Heinekein	1200	48	1248
Safari	600	24	624
Ndovu	550	22	572
Kibo gold	550	22	572
Carlsberg	800	32	832

Careful: Do not type [UNIT PRICE]+4% — Access adds 0.04, not 4% of the price.

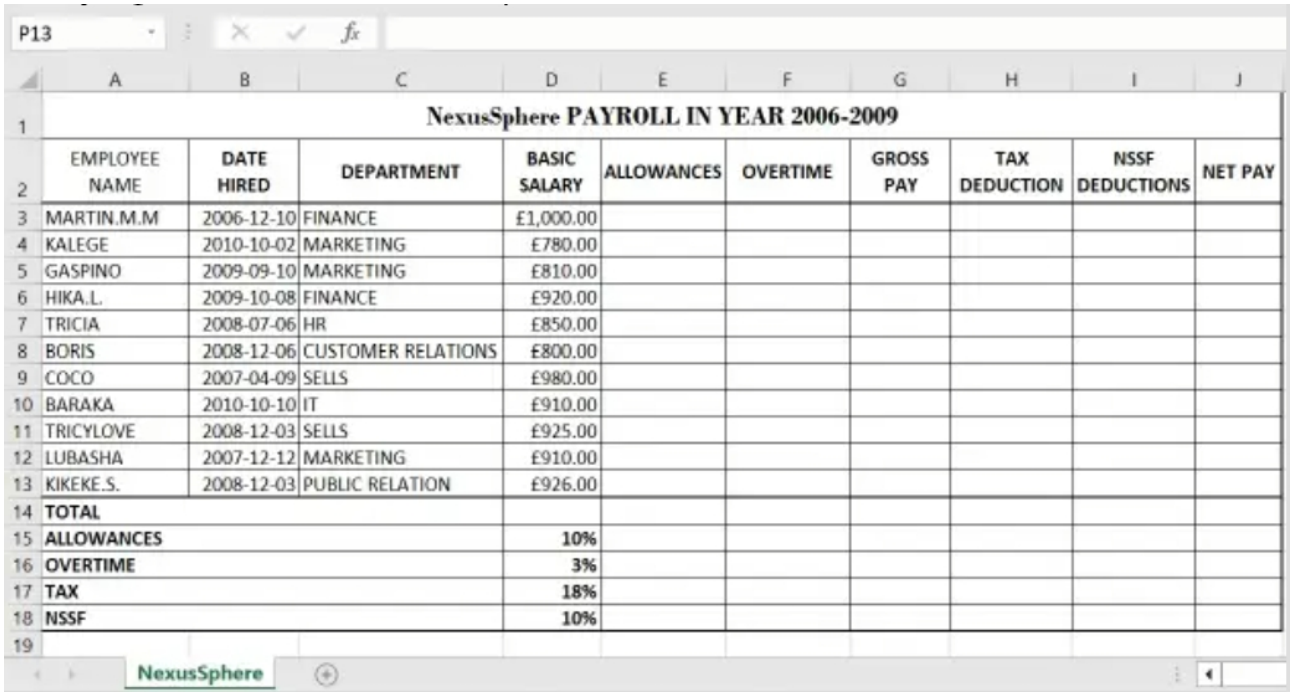
Part (j) – Print only g, h and i

1. Put your exam number on the printout: in each of the three queries add an extra column with Field `Exam: "S2911-0045"` (use your own number), then save.
2. In the left Navigation Pane double-click the query **Highest**.
3. Click **File** → **Print** → **Print Preview**. If columns do not fit, click **Landscape**.
4. Click **Print** → **OK**.
5. Do the same for **KAG&VAT** and **VAT INCLUSIVE**. Do not print other queries.

Tip: Before you finish, check the Navigation Pane shows: table PRODUCT and queries C&A, 1990 or 1996, G&L, Highest, KAG&VAT, VAT INCLUSIVE.

Question 3 – Spreadsheet with MS Excel (compulsory)

You will build the NexusSphere payroll, fill six columns with formulas, draw a chart on a new sheet, and print. Menu names are from Excel 2016–2021 / Microsoft 365.



	A	B	C	D	E	F	G	H	I	J
1	NexusSphere PAYROLL IN YEAR 2006-2009									
2	EMPLOYEE NAME	DATE HIRED	DEPARTMENT	BASIC SALARY	ALLOWANCES	OVERTIME	GROSS PAY	TAX DEDUCTION	NSSF DEDUCTIONS	NET PAY
3	MARTIN.M.M	2006-12-10	FINANCE	£1,000.00						
4	KALEGE	2010-10-02	MARKETING	£780.00						
5	GASPINO	2009-09-10	MARKETING	£810.00						
6	HIKA.L.	2009-10-08	FINANCE	£920.00						
7	TRICIA	2008-07-06	HR	£850.00						
8	BORIS	2008-12-06	CUSTOMER RELATIONS	£800.00						
9	COCO	2007-04-09	SELLS	£980.00						
10	BARAKA	2010-10-10	IT	£910.00						
11	TRICYLOVE	2008-12-03	SELLS	£925.00						
12	LUBASHA	2007-12-12	MARKETING	£910.00						
13	KIKEKE.S.	2008-12-03	PUBLIC RELATION	£926.00						
14	TOTAL									
15	ALLOWANCES			10%						
16	OVERTIME			3%						
17	TAX			18%						
18	NSSF			10%						
19										

Figure 3 – the payroll you must build

Step 0 – Type the data

1. Open **Excel** → **Blank workbook**.
2. Row 2 headings: A2 EMPLOYEE NAME, B2 DATE HIRED, C2 DEPARTMENT, D2 BASIC SALARY, E2 ALLOWANCES, F2 OVERTIME, G2 GROSS PAY, H2 TAX DEDUCTION, I2 NSSF DEDUCTIONS, J2 NET PAY.
3. Type the 11 employees in A3:D13 exactly as Figure 3.
4. Type TOTAL, ALLOWANCES, OVERTIME, TAX, NSSF in A14:A18.
5. Type the rates: D15 `10%`, D16 `3%`, D17 `18%`, D18 `10%`.
6. **File** → **Save As** → your exam-number folder → name e.g. `PAYROLL.xlsx` → **Save**.

(i) Format row A1 – the title

1. Click A1 and type `NexusSphere PAYROLL IN YEAR 2006-2009`.
2. Select **A1:J1** → **Home** → **Merge & Center**.
3. Press **Ctrl + B** for bold. Choose a font like the figure (e.g. Times New Roman, size 14).

(ii) Wrap the headings in row 2

1. Select **A2:J2** → **Home** → **Wrap Text**.
2. Optional: click **Middle Align** and **Center** to match the figure.

(iii) Rename the sheet and colour the tab red

1. Double-click the **Sheet1** tab at the bottom → type `NexusSphere` → **Enter**.
2. Right-click the tab → **Tab Color** → **Red**.

(iv) Show the pound symbol automatically

1. Select **D3:J13**.
2. Press **Ctrl + 1** → **Number** tab → **Currency**.
3. **Symbol:** choose **£ English (United Kingdom)**; **Decimal places:** 2 → **OK**.

Careful: Do not type the £ sign yourself. Typed symbols turn numbers into text and formulas stop working.

(v) Add borders

1. Select **A1:J18** → **Home** → Borders arrow → **All Borders**.

Formulas for (vi) to (xi)

Type each formula in row 3, press Enter, then copy it down: click the cell, then drag the small square at its bottom-right corner (the **fill handle**) down to row 13.

Tip: An **absolute reference** such as D15 does not change when you copy the formula down. After clicking D15 in a formula, press **F4** to add the $ signs.

Part	Click cell	Type this formula	Copy down to	Row 3 shows
(vi) Allowances 10%	E3	=D3*D15	E13	£100.00
(vii) Overtime 3%	F3	=D3*D16	F13	£30.00
(viii) Gross pay	G3	=SUM(D3:F3)	G13	£1,130.00
(ix) Tax 18%	H3	=G3*D17	H13	£203.40
(x) NSSF 10%	I3	=G3*D18	I13	£113.00
(xi) Net pay	J3	=G3-H3-I3	J13	£813.60

You may also use the PRODUCT function, e.g. `=PRODUCT(D3,$D$15)`, as the paper's NOTE allows.

Optional totals in row 14: in D14 type `=SUM(D3:D13)` and drag right to J14.

Check your finished table

Employee	Basic	Allow.	Overtime	Gross	Tax	NSSF	Net pay
MARTIN.M.M	£1,000.00	£100.00	£30.00	£1,130.00	£203.40	£113.00	£813.60
KALEGE	£780.00	£78.00	£23.40	£881.40	£158.65	£88.14	£634.61
GASPINO	£810.00	£81.00	£24.30	£915.30	£164.75	£91.53	£659.02
HIKA.L.	£920.00	£92.00	£27.60	£1,039.60	£187.13	£103.96	£748.51
TRICIA	£850.00	£85.00	£25.50	£960.50	£172.89	£96.05	£691.56
BORIS	£800.00	£80.00	£24.00	£904.00	£162.72	£90.40	£650.88
COCO	£980.00	£98.00	£29.40	£1,107.40	£199.33	£110.74	£797.33
BARAKA	£910.00	£91.00	£27.30	£1,028.30	£185.09	£102.83	£740.38
TRICYLOVE	£925.00	£92.50	£27.75	£1,045.25	£188.14	£104.53	£752.58
LUBASHA	£910.00	£91.00	£27.30	£1,028.30	£185.09	£102.83	£740.38
KIKEKE.S.	£926.00	£92.60	£27.78	£1,046.38	£188.35	£104.64	£753.39
TOTAL	**£9,811.00**	**£981.10**	**£294.33**	**£11,086.43**	**£1,995.56**	**£1,108.64**	**£7,982.23**

Tip: Quick check: Net pay is always Gross pay × 0.72. A difference of 1p in totals is only rounding.

(xii) Draw the Stacked Line with Markers chart

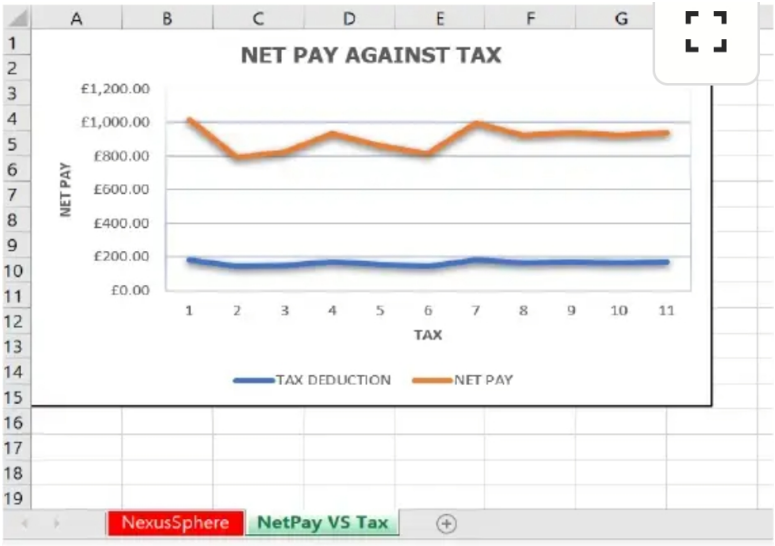


Figure 4 – the chart you must make

1. Select **H2:H13** (TAX DEDUCTION). Hold **Ctrl** and select **J2:J13** (NET PAY).
2. Click **Insert** → the **Line** chart icon → **Stacked Line with Markers**.
3. Click **Chart Design** → **Move Chart** → **New sheet** → type `NetPay VS Tax` → **OK**.
4. Click the chart title, delete it and type `NET PAY AGAINST TAX`.
5. Select the title text → **Home** → font **Tahoma**, size **12**.

(xiii) Style 13, tab colour and chart elements

1. Click the chart → **Chart Design** → open the **Chart Styles** gallery → point at each style until the tip says **Style 13**, then click it.
2. Right-click the **NetPay VS Tax** tab → **Tab Color** → **Green**.

3. **Chart Design → Add Chart Element → Axis Titles → Primary Vertical**; type `NET PAY`.
4. **Add Chart Element → Axis Titles → Primary Horizontal**; type `TAX`.
5. **Add Chart Element → Legend → Bottom**. Make sure gridlines show, as in Figure 4.

Tip: In a stacked line chart the Net Pay line sits on top of the Tax line, so it shows about £1,000 even though real net pay is about £800. That is normal.

(xiv) Black solid border on the chart

1. Right-click the outer edge of the chart → **Format Chart Area**.
2. Click the paint-bucket icon (**Fill & Line**) → **Border** → **Solid line**.
3. **Color** → **Black**. Close the panel.

(xv) Print – table landscape, chart portrait

1. Add your exam number on both sheets: **Insert → Header & Footer** → type the number (or **Page Layout** → Page Setup arrow → **Header/Footer → Custom Header**).
2. Go to the **NexusSphere** sheet → **Page Layout → Orientation → Landscape**.
3. **File → Print**; check the whole table fits on one page → **Print**.
4. Go to the **NetPay VS Tax** sheet → **Page Layout → Orientation → Portrait**.
5. **File → Print → Print**.
6. Press **Ctrl + S** one last time. Check the printouts match your saved file.

Final checklist before you hand in

- ☐ Title merged and bold across A1:J1
- ☐ Row 2 headings wrapped
- ☐ Sheet named NexusSphere with a red tab
- ☐ Pound format on D3:J13
- ☐ Borders on A1:J18
- ☐ Allowance, Overtime and NSSF formulas use D15, D16, D18
- ☐ J3 shows £813.60
- ☐ Stacked line with markers chart on its own sheet, Tahoma 12 title
- ☐ Style 13, green tab named NetPay VS Tax, axis titles and legend
- ☐ Black solid chart border
- ☐ Table printed landscape, chart printed portrait
- ☐ Exam number on every page; everything saved in your exam-number folder